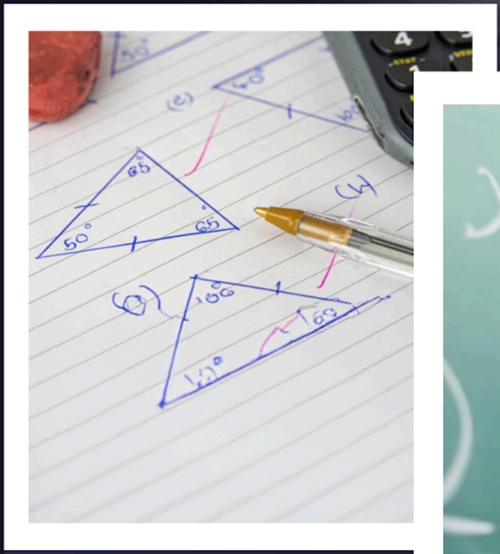
Quantitative Aptitude

Number System

STUDY NOTES

Factors



Factor

Definition:

A **factor** (or divisor) is a number that divides another number exactly, leaving no remainder. In mathematical terms, if a number a divides another number b perfectly, then a is considered a factor of b. This relationship can be expressed as:

$$b = a \times n$$

Where n is an integer, both a and n multiply to give b without any remainder.

Types of Factors:

1. **Positive Factors:** These are positive integers that divide the number without a remainder.
2. **Negative Factors:** If a positive number has a factor, then the negative of that factor will also divide the number. For example, for 6, both -2 and -3 are negative factors since $(-2) \times (-3) = 6$.

Properties of Factors:

The factors of N are always smaller than or equal to N.

- **Factors of a Prime Number:** A prime number has exactly two factors: 1 and itself. For example, 7 has factors 1 and 7 only.
- **Factors of a Composite Number:** A composite number has more than two factors. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12.
- **1 as a Factor:** 1 is a universal factor because it divides every number.
- **Self Factor:** Every number is a factor of itself since it divides itself exactly once.

Factor Pairs:

- Factors often come in pairs. For example, if $a \times b = N$, then a and b are factor pairs of N.
- For example, for 12, the factor pairs are (1, 12), (2, 6), and (3, 4) since $1 \times 12 = 12$, $2 \times 6 = 12$, and $3 \times 4 = 12$.

Example 1: For the number 18, the factors are 1, 2, 3, 6, 9, and 18.

This is because:

$1 \times 18 = 18$

$2 \times 9 = 18$

$3 \times 6 = 18$

Factors can be both positive and negative. For instance, -1, -2, -3, -6, -9, and -18 are also factors of 18.

How to calculate the number of factors

1. Calculate the total number of factors:

To calculate the total number of factors of a given number, we use the prime factorization method. This method provides a systematic way to count all divisors of a number, including both prime and composite factors.

Find the prime factorization: Express the number as a product of prime factors. If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \dots p_k^{ek}$,

Where p_1, p_2, p_3 p_k are prime numbers and e1, e2 and ek are their respective power.

Then, Total number of factors T(N) is:

$$\text{Total Number of Factors} = (e_1 + 1) \times (e_2 + 1) \times (e_3 + 1) \times \dots (e_k + 1)$$

This includes 1 and the number itself.

Example 2: The total number of factors for number 180 are

Solution:

Concept:

Let us assume N is a natural number, for which we need to find the factors.

If we convert N into the product of prime numbers by the prime factorization method, we can represent it as;

$N = X^a \times Y^b \times Z^c$

Where X, Y, and Z are the prime numbers and a, b, and c are their respective powers.

Now, the formula for the total number of factors for a given number is given by;

Total Number of Factors for N = (a + 1) (b + 1) (c + 1)

Calculation:

Given number is 180;

By prime factorization, 180 can be expressed as

$180 = 3^2 \times 2^2 \times 5^1$

Total number of factors will be

N = (2 + 1)(2 + 1)(1 + 1) = 18

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2. Calculate the total number of odd factors :

Find the prime factorization: Express the number as a product of prime factors. If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \dots p_k^{ek}$,

Where p_1, p_2, p_3 pk are prime numbers and e1, e2, and ek are their respective power.

Exclude the Even Prime Factor:

Odd factors do not include the prime factor 2. Therefore, in the prime factorization, ignore any factors of 2.

Then, Total number of odd factors is:

$$\text{Total Number of Factors} = (e_1 + 1) \times (e_2 + 1) \times (e_3 + 1) \times \ldots (e_k + 1)$$

Example 3: Find the number of odd factors of 7500.

Solution:

Given:

The number is 7500.

Concept used:

If a number $N = x^a . y^b . z^c$ where x, y, z are distinct prime numbers and a, b, c are natural numbers,

Number of odd factors = $(a + 1) \times (b + 1) \times (c + 1)$

If $x = 2$ (only even prime number), then

Number of odd factors = $(b + 1) \times (c + 1)$

Explanation:

$7500 = 2^2 \times 3^1 \times 5^4$

$\Rightarrow$ Number of odd factors = $(1 + 1) \times (4 + 1)$

$\Rightarrow$ Number of odd factors = 10

∴ The number of odd factors is 10.

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3. Calculate total number of even factors :

Find the prime factorization: Express the number as a product of prime factors. If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \ldots p_k^{ek}$,

Where p_1, p_2, p_3, p_k are prime numbers and e1, e2, and ek are their respective power.

Then, Total number of even factors is:

Total Even Factors = Total Factors – Total Odd Factors

Total Number of even factors = $e_1 \times (e_2 + 1) \times (e_3 + 1) \times \dots (e_k + 1)$

Example 4: The number of even factors of 81900 will be

Solution:

Given:

The number = 81900

Concept used:

Prime factors of number = $2^p \times 3^q \times 5^r \times 7^s \times$ ....

Number of even factors = $p \times (q + 1)(r + 1)(s + 1)$ .....

Calculations:

Prime factors of 81900 = $5 \times 5 \times 2 \times 2 \times 3 \times 3 \times 7 \times 13$

$\Rightarrow 2^2 \times 3^2 \times 5^2 \times 7^1 \times 13^1$

Number of even factors = $2 \times (2 + 1)(2 + 1)(1 + 1)(1 + 1)$

$\Rightarrow 2 \times 3 \times 3 \times 2 \times 2 = 72$

∴ The number of even factors of 81900 is 72

How to calculate the sum of factors

1. Calculate the total sum of factors:

Find the prime factorization: Express the number as a product of prime factors. If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \dots p_k^{ek}$, Where p_1, p_2, p_3, p_k are prime numbers and e1, e2, and ek are their respective power.

Then the sum of all factors S(N) is given by:

$$S(N) = (1 + p_1 + p_1^2 + \cdots + p_1^{e1})(1 + p_2 + p_2^2 + \cdots + p_2^{e2}) \ldots (1 + p_k + p_k^2 + \cdots + p_k^{ek})$$

This formula works because each term in the product represents the sum of factors contributed by each prime factor raised to different powers.

Example 5: Find the sum of factors of 90.

Solution:

Concept used:

$N = p^a q^b r^c$

Where, p, q, and r are prime factors of number N.

a, b, and c are non-negative powers.

Sum of factors = $(p^0 + p^1 + p^2 \ldots\ldots\ldots + p^n)\ (q^0 + q^1 + q^2 \ldots\ldots\ldots q^n)\ (r^0 + r^1 + r^2 \ldots\ldots r^n)$

Calculations:

⇒ $90 = 2 \times 3 \times 3 \times 5$

⇒ $90 = 2^1 \times 3^2 \times 5^1$

⇒ Sum of factors = $(2^0 + 2^1) \times (3^0 + 3^1 + 3^2) \times (5^0 + 5^1)$

⇒ Sum of factors = $(1 + 2) \times (1 + 3 + 9) \times (1 + 5)$

⇒ Sum of factors = $3 \times 13 \times 6$

⇒ Sum of factors = 234

∴ The sum of factors of 90 is 234.

2. Calculate total sum of odd factor :

Find the prime factorization: Express the number as a product of prime factors. If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \dots p_k^{ek}$,

Where p_1, p_2, p_3, p_k are prime numbers and e1, e2, and ek are their respective powers.

Exclude the Even Prime Factor:

Odd factors do not include the prime factor 2. Therefore, in the prime factorization, ignore any factors of 2.

Then the sum of all odd factors $S_{odd}(N)$ is given by:

$$S_{odd}(N) = (1 + p_1 + p_1^2 + \dots + p_1^{e1})(1 + p_2 + p_2^2 + \dots + p_2^{e2}) \dots (1 + p_k + p_k^2 + \dots + p_k^{ek})$$

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3. Calculate total sum of even factor :

Find the prime factorization: Express the number as a product of prime factors. If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \dots p_k^{ek}$,

Where p_1, p_2, p_3, p_k are prime numbers and e1, e2, and ek are their respective power.

Then the sum of all even factors $S_{even}(N)$ is given by:

Total Sum of Even Factors = Sum of factors - Sum of odd factors

$$S_{even}(N) = (2 + 2^2 + \dots + 2^{e0})(1 + p_1 + p_1^2 + \dots + p_1^{e1}) \dots (1 + p_k + p_k^2 + \dots + p_k^{ek})$$

Here, 2^{e0} represents the power of 2 in the prime factorization.

Example 6: Total Sum of Even Factors of 180

Solution:

1. **Prime Factorization:**
   - $180 = 2^2 \times 3^2 \times 5^1$
2. **Separate the Power of 2:**
   - The power of 2 in 180 is 2^2. So the sum of all powers of 2 (excluding 1) is: $2 + 2^2 = 2 + 4 = 6$
3. **Calculate the Sum of Factors of the Remaining Part (Odd Portion):**
   - The remaining part after factoring out 2^2 is $3^2 \times 5^1$.
   - Sum of factors of 3^2: $1 + 3 + 3^2 = 1 + 3 + 9 = 13$
   - Sum of factors of 5^1: $1 + 5 = 6$
4. **Multiply the Results:**
   $S_{even}(180) = 6 \times 13 \times 6 = 468$

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4. Sum of reciprocal of all factor :

Calculate the Sum of All Factors using the sum of factors formula.

Divide the Sum by the Number to find the sum of the reciprocals of all factors:

Sum of reciprocals of all factors = $\frac{Total\ Sum\ of\ all\ factors}{Number} = \frac{S(N)}{N}$

Example 7: What is the sum of reciprocal of all factors of number 900

Solution:

Given:

900

Formula used:

Sum of reciprocal of all factors = Sum of all factors/number

Concept used:

If $k = a^x \times b^y$, then

Sum of all factors = $(a^0 + a^1 + a^2 + ... + a^x)(b^0 + b^1 + b^2 + ... + b^y)$

Calculation:

$900 = 2^2 \times 3^2 \times 5^2$

Sum of all factors = $(2^0 + 2^1 + 2^2)(3^0 + 3^1 + 3^2)(5^0 + 5^1 + 5^2)$

$\Rightarrow (1 + 2 + 4)(1 + 3 + 9)(1 + 5 + 25)$

$\Rightarrow 7 \times 13 \times 31$

$\Rightarrow 2821$

Sum of reciprocal of all factors = 2821/900

$\Rightarrow 3.13$

∴ **Required answer is 3.13**

How to calculate product of factor

Find the prime factorization: Express the number as a product of prime factors. If the prime factorization of a number N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} ... p_k^{ek}$,

Where p_1, p_2, p_3 p_k are prime numbers and e1, e2 and ek are their respective powers. Then,

Total number of factors T(N) is:

Total Number of Factors = $(e_1 + 1) \times (e_2 + 1) \times (e_3 + 1) \times ... (e_k + 1)$

Product of all factors = $N^{T(N)/2}$

Example 8: Find the number of factors and the products of all factors of 1224?

Solution:

Formula Used:

$N = A^p \times B^q \times C^r$ .Here A, B, and C are prime numbers, and p, q, and r were respective powers of those prime numbers.

Total numbers of factors for " N "= (p + 1)(q +1)(r +1)

Product of all factors of "N" = $(N)^{\text{Total no. of factors}/2}$

Calculation:

By using the above formula, we get

⇒ Number of factors for 1224 = 8 × 17 × 9 = $2^3 \times 3^2 \times 17$

⇒ Total numbers of factors for 1224 = (3 +1)(2 + 1)(1 +1) = 4 × 3 × 2

⇒ Total numbers of factors for 1224 = 24

Now the product of all factors of 1224 = $(1224)^{24/2} = (1224)^{12}$

∴ Total numbers of factors for 1224 = 24 and product is $(1224)^{12}$

How to calculate total perfect square and perfect cube factors

1. Calculate total number of perfect square factors:

To calculate the **total number of perfect square factors** of a given number, we need to focus on the nature of perfect squares. A perfect square factor must have even exponents for all prime factors in its prime factorization. This means that in the prime factorization of a number, only those factors that can be formed using even exponents of each prime factor are considered perfect square factors.

Identify Conditions for Perfect Square Factors:

- A factor is a perfect square if and only if all exponents in its prime factorization are even. This restriction ensures that the factor is a perfect square.

- For each prime factor p_i^{ei} in the prime factorization of N, we can use even exponents from 0 up to the highest even exponent less than or equal to e_i.

Apply the Formula:

Multiply the number of choices for each prime factor to find the total number of perfect square factors.

Formula: If the prime factorization of N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \dots p_k^{ek}$,

Where p_1, p_2, p_3 p_k are prime numbers and e1, e2 and ek are their respective power.

Then the total number of perfect square factors of N is:

Total perfect square factors = $(e_1/2 + 1) \times (e_2/2 + 1) \times \cdots \times (e_k/2 + 1)$

Example 9: Perfect Square Factors of 180

Solution:

1. **Prime Factorization**:
   - $180 = 2^2 \times 3^2 \times 5^1$
2. **Identify the Even Exponents for Perfect Square Factors**:
   - For 2^2: Possible exponents for a perfect square factor are 0 and 2, giving (2/2) + 1 = 2 choices.
   - For 3^2: Possible exponents for a perfect square factor are 0 and 2, giving (2/2) + 1 = 2 choices.
   - For 5^1: The exponent is odd, so we can only use 5^0 (no factor of 5) to keep the factor a perfect square. This provides (1+1)/2 = 1 choice.
3. **Calculate the Total Number of Perfect Square Factors**:
   Total perfect square factors of 180 = 2 × 2 × 1 = 4

2. Calculate total number of perfect cube factors :

To calculate the **total number of perfect cube factors** of a given number, we use its prime factorization. A factor is a perfect cube if all exponents in its prime factorization are multiples of 3. This restriction ensures that each factor formed is a perfect cube.

Steps to Calculate the Total Number of Perfect Cube Factors

1. **Prime Factorization**:
   - Start by expressing the number as a product of its prime factors.
2. **Identify the Conditions for Perfect Cube Factors**:
   - A factor is a perfect cube if and only if all exponents in its prime factorization are multiples of 3 (e.g., 0, 3, 6, etc.).
   - This restriction limits the possible exponents for each prime factor to 0, 3, 6, and so on, up to the maximum exponent of that prime factor in the prime factorization of the number.
3. **Calculate the Number of Choices for Each Prime Factor**:
   - For each prime factor with an exponent e, the possible exponents for perfect cube factors are multiples of 3 from 0 up to the largest multiple of 3 that does not exceed e.
   - Therefore:
     - If e is a multiple of 3, the possible choices are 0,3,6,…,e which gives $(e^3/3) + 1$ choices.
     - If e is not a multiple of 3, round down to the nearest multiple of 3, say e′, and the choices are 0,3,6,…,e′, which gives $(e'/3) + 1$ choices.
4. **Apply the Formula**:
   - Multiply the number of choices for each prime factor to find the total number of perfect square factors.

Formula: If the prime factorization of N is

$N = p_1^{e1} \times p_2^{e2} \times p_3^{e3} \dots p_k^{ek}$,

Where p_1, p_2, p_3 p_k are prime numbers and e1, e2 and ek are their respective powers.

Then the total number of perfect cube factors of N is:

Total perfect square factors = $(e_1/3 + 1) \times (e_2/3 + 1) \times \cdots \times (e_k/3 + 1)$

Proper factor

Proper factors of a number are all the factors of that number except 1 and the number itself.

- A proper factor of a number N is any positive integer divisor of N that is less than N.

Let the total number of factors of a number be 'n' then:

Total number of proper factors = n - 2

Example: Consider the number 12:

The factors of 12 are: 1, 2, 3, 4, 6, 12.

Total number of factors = 6

The proper factors of 12 are 2, 3, 4, 6.

Total number of proper factors = n - 2 = 6 - 2 = 4
